

capstoneproject

May 24, 2026

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[8]: #Importing required libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report

#Loading a CSV file
data = pd.read_csv(r'ai_displacement.csv')

#Data cleaning
data.drop_duplicates(inplace=True)
data.fillna(method='ffill',inplace=True)

#Converting dates
data['year'] = pd.to_datetime(data['year'],format='%Y')

#Sector-wise layoffs
print("1. Sector-wise layoffs: \n")
sector_layoffs = data.
    ↳groupby('industry_sector')['ai_cited_layoff_announcements'].sum().
    ↳sort_values(ascending = False)
print("Top 3 sectors facing highest layoffs: ")
print(sector_layoffs.head(3))
print('\n')
plt.figure(figsize=(12,6))
sector_layoffs.plot(kind='bar')
plt.title('Sector-wise layoffs')
plt.xlabel('Sector')
plt.ylabel('Layoffs')
plt.show()

#Country-wise displacement
print("2. Country-wise displacement: \n")
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country_displacement = data.
    ↳groupby('country')['pct_sector_workforce_displaced'].mean().
    ↳sort_values(ascending=False)
print("Top 3 countries with high displacement rates: ")
print(country_displacement.head(3))
print("\n")
plt.figure(figsize=(12,6))
country_displacement.plot(kind='bar')
plt.title('Country-wise Displacement Rate')
plt.xlabel('Country')
plt.ylabel('Displacement Rate')
plt.show()

#Female dominated sectors
print("3. Female dominated sectors: \n")
female_dominance = data[data['pct_workforce_female']*100>50].
    ↳groupby('industry_sector')['pct_workforce_female'].mean().
    ↳sort_values(ascending=False)
print("Sectors with high female workers:")
print(female_dominance.head())
print('\n')
colors = ['lightblue','lightgreen','red','gold']
plt.figure(figsize=(8,8))
female_dominance.plot(kind='pie', autopct='%1.1f%%',colors=colors)
plt.title('Female Dominated Sectors')
plt.ylabel('')
plt.show()

#Reskilling vs Displacement analysis
print("4. Reskilling vs Displacement Analysis: \n")
plt.figure(figsize=(10,6))
sns.
    ↳scatterplot(data=data,x='reskilling_programs_count',y='pct_sector_workforce_displaced',hue=
plt.title('Reskilling vs Displacement')
plt.xlabel('Number of Reskilling Programs')
plt.ylabel('Workforce Displaced')
plt.show()

#AI Wage Premium analysis
print("5. AI Wage Premium Analysis: \n")
wage_trend = data.groupby(data['year'].dt.year)['ai_skill_wage_premium_pct'].
    ↳mean()
plt.figure(figsize=(10,6))
wage_trend.plot(marker='o')
plt.title('AI Skill Wage Premium Trend')
plt.xlabel('Year')
plt.ylabel('Wage Premium %')

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plt.grid(True)
plt.show()

#Net Workforce Change
print("6. Net Workforce Change: \n")
net_change = data.groupby(data['year'].dt.year)['net_workforce_change_pct'].
    ↪sum()
plt.figure(figsize=(10,6))
net_change.plot(marker='o')
plt.title('Net Workforce Change')
plt.xlabel('Year')
plt.ylabel('Net Change')
plt.grid(True)
plt.show()

#Risk Classification
def classify_risk(score):
    if score*100 >= 80:
        return 'High'
    elif score*100 >= 50:
        return 'Medium'
    else:
        return 'Low'
data['risk_level'] = data['sector_automation_risk_score'].apply(classify_risk)

#ML Model
print("7. Machine Learning Model: \n")
le = LabelEncoder()
categorical_cols = ['industry_sector', 'country', 'income_group']
for col in categorical_cols:
    data[col] = le.fit_transform(data[col])
X = data[[
    'industry_sector',
    'country',
    'ai_adoption_index',
    'sector_automation_risk_score',
    'reskilling_programs_count',
    'pct_workforce_female'
]]

y = data['risk_level']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
    ↪random_state=42)
model = RandomForestClassifier()
model.fit(X_train, y_train)
predictions = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, predictions))

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print(classification_report(y_test, predictions))
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1. Sector-wise layoffs:

Top 3 sectors facing highest layoffs:

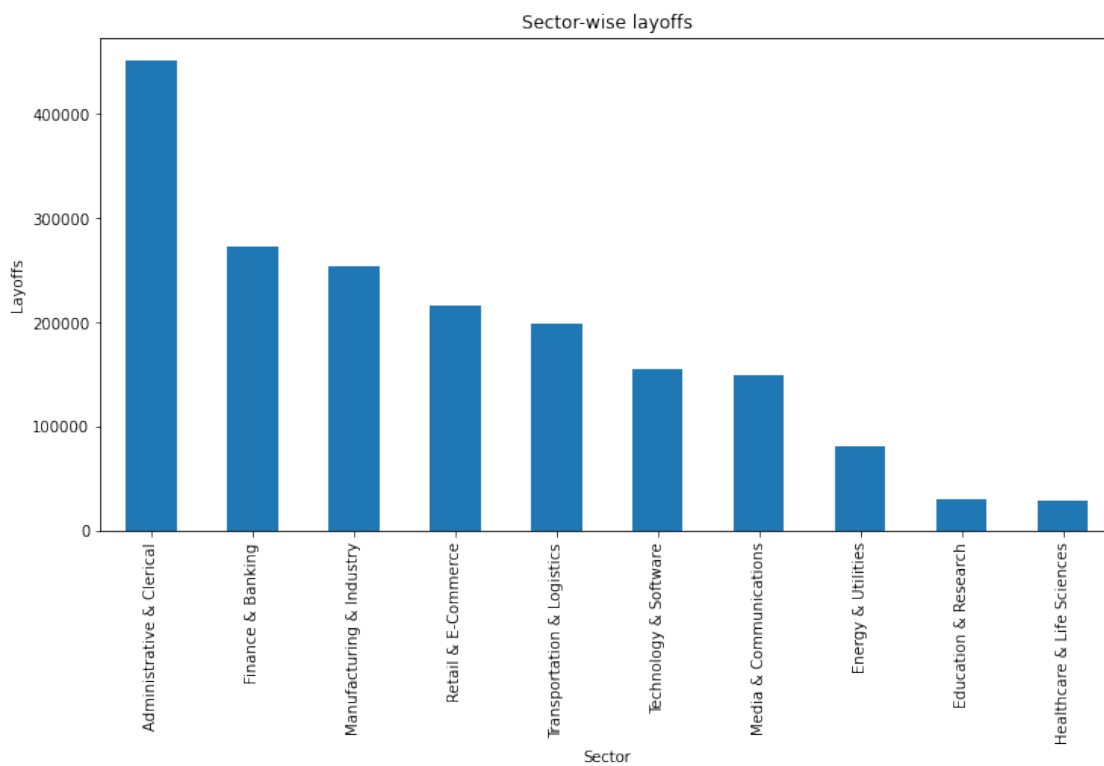
industry_sector

Administrative & Clerical 450999

Finance & Banking 273538

Manufacturing & Industry 253862

Name: ai_cited_layoff_announcements, dtype: int64



2. Country-wise displacement:

Top 3 countries with high displacement rates:

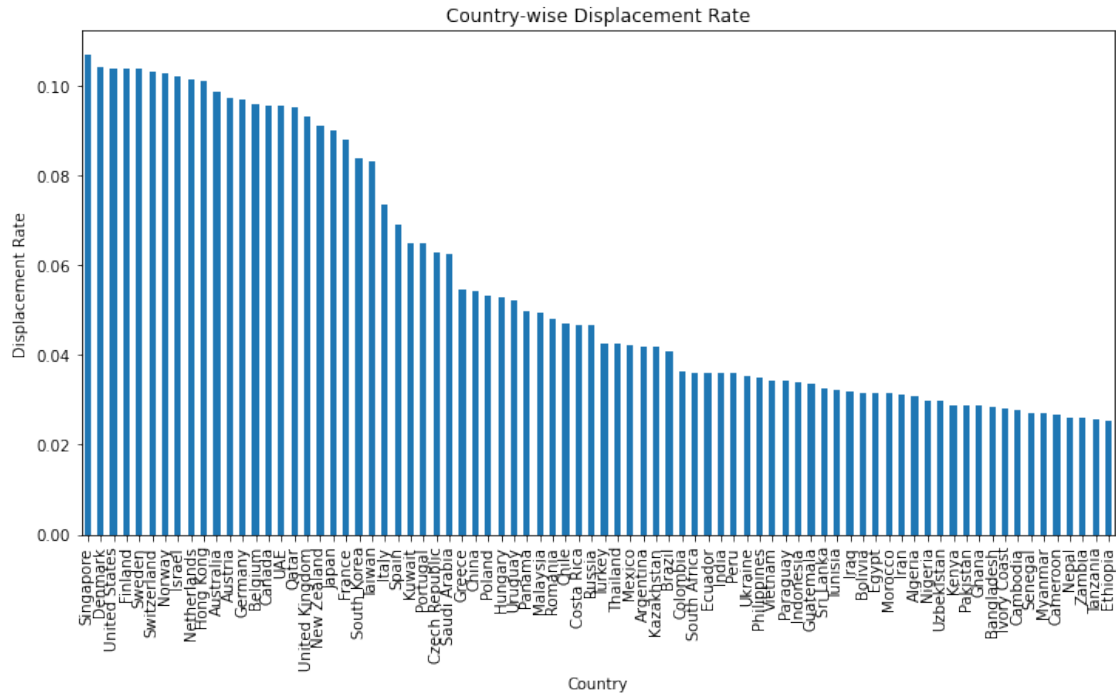
country

Singapore 0.106867

Denmark 0.104123

United States 0.103910

Name: pct_sector_workforce_displaced, dtype: float64



3. Female dominated sectors:

Sectors with high female workers:

industry_sector

Administrative & Clerical 0.79

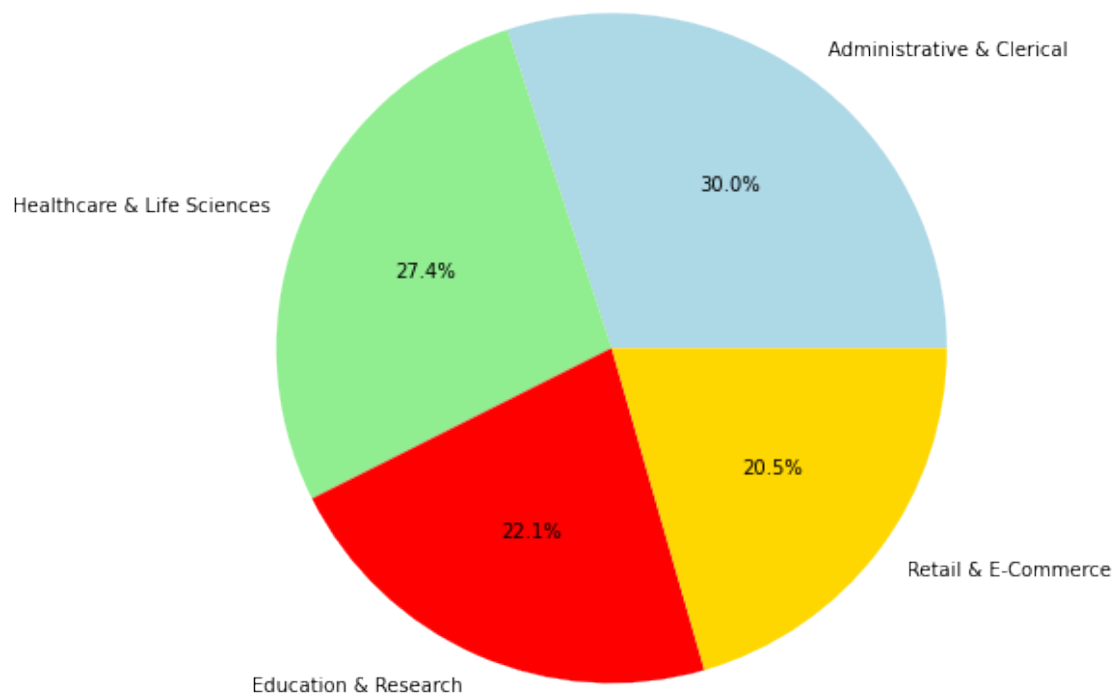
Healthcare & Life Sciences 0.72

Education & Research 0.58

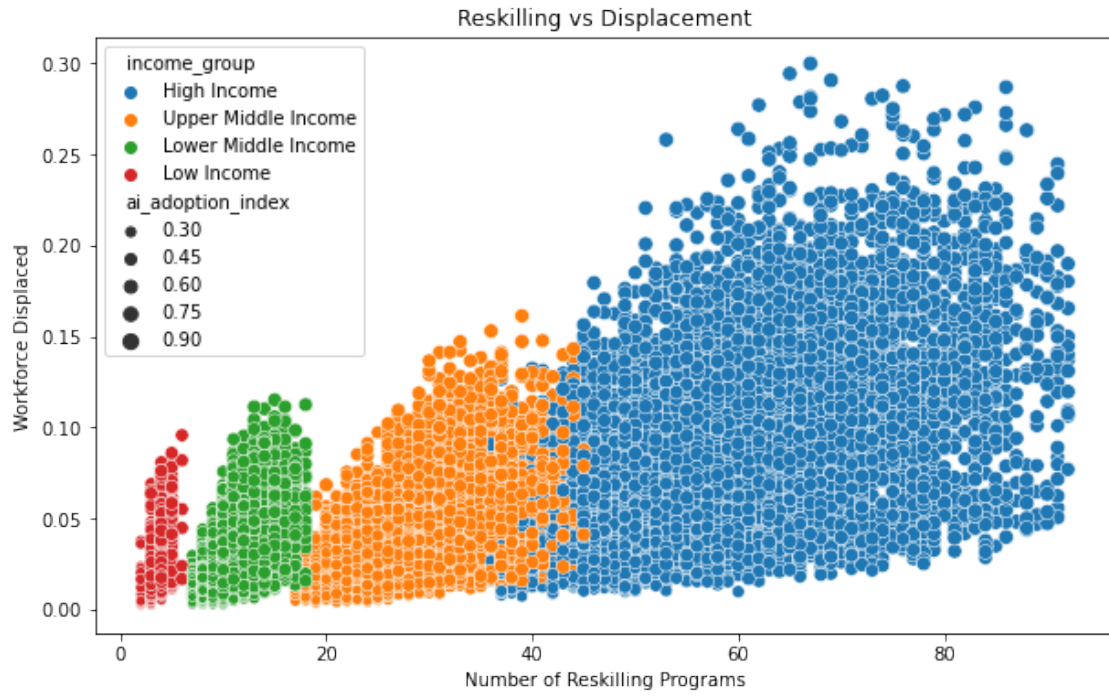
Retail & E-Commerce 0.54

Name: pct_workforce_female, dtype: float64

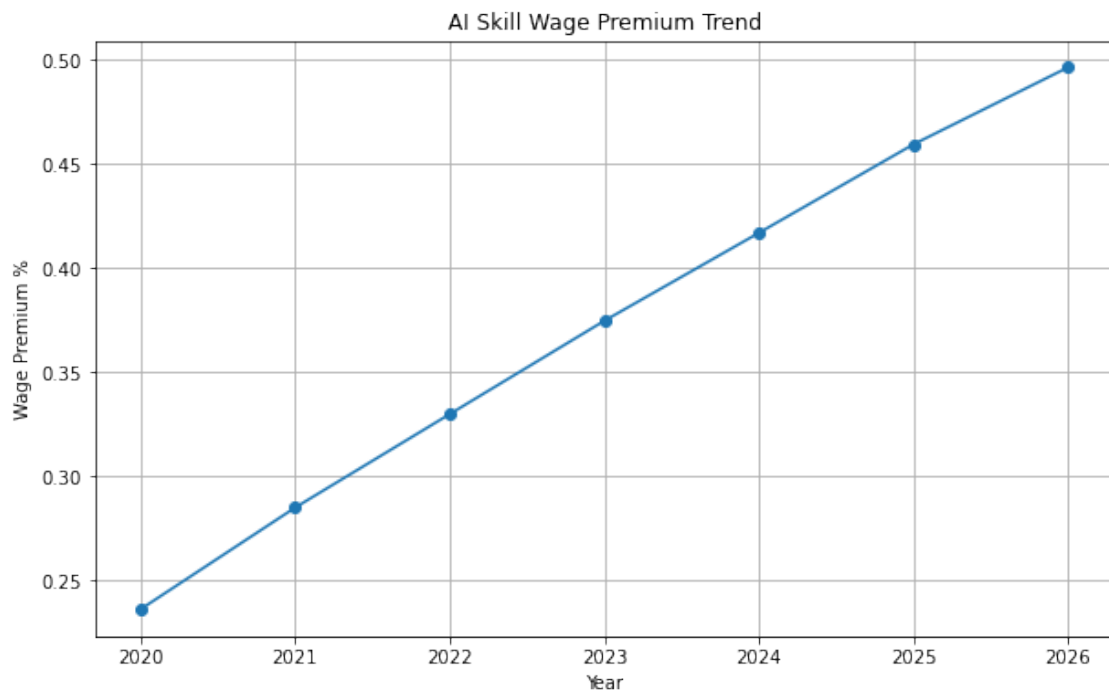
Female Dominated Sectors



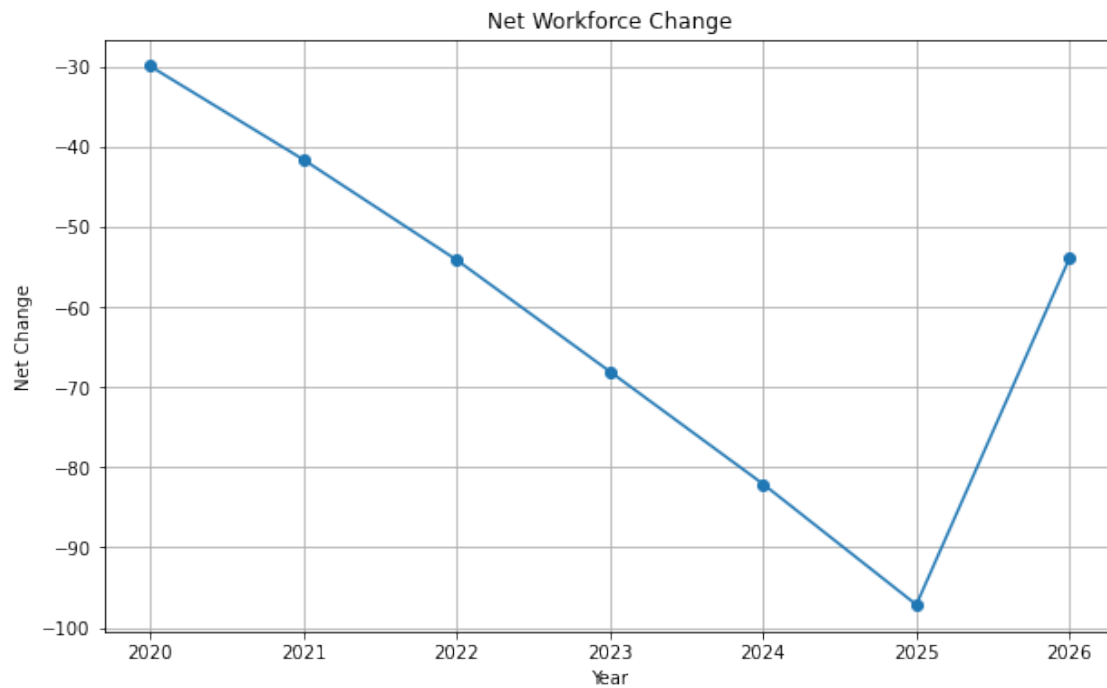
4. Reskilling vs Displacement Analysis:



5. AI Wage Premium Analysis:



6. Net Workforce Change:



7. Machine Learning Model:

Accuracy: 1.0

	precision	recall	f1-score	support
High	1.00	1.00	1.00	136
Low	1.00	1.00	1.00	1711
Medium	1.00	1.00	1.00	2313
accuracy			1.00	4160
macro avg	1.00	1.00	1.00	4160
weighted avg	1.00	1.00	1.00	4160

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